

Pastur's theorem

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We've seen that for $H = -\Delta + V$ acting on $L^2(\mathbb{Z}^d)$, where V is id $\mathbb{P}$, with compact supp, almost surely $\text{Spec}(H) = [0, 4d] + \text{supp}(P)$

One can actually show: $\text{Spec}^{\text{pp}}(H)$, $\text{Spec}^{\text{ac}}(H)$, $\text{Spec}^{\text{sc}}(H)$ are all deterministic.

Proof. Let $T_x H = -\Delta + T_x V$, where $T_x V(y) = V(x+y)$. for some $x \in \mathbb{Z}^d$.

Then $\text{Spec}^{\#}(T_x H) = \text{Spec}^{\#}(H)$, almost surely, since $T_x H = T_x \circ H \circ T_x^{-1}$.
($\# = \text{pp}, \text{ac}, \text{sc}$)

$\Rightarrow$ For any open interval I , the event $I \cap \text{Spec}^{\#}(H) \neq \emptyset$ (on the space of $(\mathbb{R}^d, \mathbb{P}^d)$) is invariant

under the operations $(T_x)_{x \in \mathbb{Z}^d}$. Therefore its probability $\in \{0, 1\}$

$\Rightarrow$ By taking $I = (p, q)$ for all $p, q \in \mathbb{Q}$, $p < q$, we get that $\text{Spec}^{\#}(H)$ is deterministic.

• More generally, this holds for all ergodic operator

Let (Ω, μ) be a probability space, and $(T_x)_{x \in I}$ an ergodic family of measure-preserving transformations, i.e., for any event E s.t. $T_x E = E$, for all $x \in I$, we must have $\mu(E) \in \{0, 1\}$.

A random operator H defined over this prob space is an ergodic operator, if for every $x \in I$, $w \in \mathbb{R}$, we have
(acting on Hilbert space $\mathcal{H}$)
 $H(T_x w)$ is unitarily equivalent to $H(w)$.

[$H(T_x w) = U^{\dagger} H(w) U$ for some unitary operator U]

(Pastur) Any ergodic operator has deterministic $\text{Spec}^{\#}$ ($\# = \text{pp}, \text{ac}, \text{sc}$)

Proof. Same as above, in the more abstract language. Use that $\text{Spec}^{\#}(H) = \text{Spec}^{\#}(U^{\dagger} H U)$ for unitary U .

Another (less obvious) example.

almost Mathieu operator $H^{\lambda, \alpha}$ acting $L^2(\mathbb{Z})$, via $H^{\lambda, \alpha} f(x) = f(x+1) + f(x-1) + 2\lambda \cos(2\pi(u + \alpha x)) f(x)$
"randomness": $u \sim \text{unif}[0, 1]$

• α rational: periodic, equiv to finite dim problem.

• α irrational: ergodic operator. $\Rightarrow$ spec & decomposition deterministic

(Ten Martini Problem: for all $\lambda \neq 0$, irrational α , and all u , the spectrum is a Cantor set)

(Asked by Mark Kac, who offered ten Martini for solving it; also Barry Simon)

(Avila - Jitomirskaya, 05')

$\forall \alpha$ irrational:

• $\lambda > 1$: $\text{Spec}(H^{\lambda, \alpha})$ is only pure point (Jitomirskaya, 99')

• $\lambda = 1$: $\text{Spec}(H^{\lambda, \alpha})$ is only singular continuous (Gordon, Jitomirskaya, Last, Simon, 91')

• $0 < \lambda < 1$: $\text{Spec}(H^{\lambda, \alpha})$ is only absolute continuous (Avila, 08')

(Hofstadter's butterfly, as α varies)